# ASSIGNMENT

1. The molecules of ice, water or steam always contain:
   1) Two atoms of oxygen. 2) Two atoms of hydrogen.
   3) One atom of hydrogen. 4)3 atom of oxygen.
2. When we add common salt to water, salt solution is formed.
   Which of the following changes might have occurred in the salt solution?
   1) The chemical composition of molecules changed slightly.
   2) The mass of the salt solution increased.
   3) Salt solution has some new substances.
   4) The chemical composition remained the same.
3. Melting of butter is not a:
   1) Fast change 2) Desirable change
   3) Reversible change 4) Permanent change
4. Statement (A) : Water changes to steam by absorbing a certain amount of energy.
   Statement (B): The heat energy supplied during the above change is utilised to change the composition of the molecules of a substance.
   1) Statement 'A' is true. 2) Statement 'B' is false.
   3) Statements 'A' and 'B' are both true.
   4) Statement 'A' is false but 'B' is true.
5. The heating of iron fillings and sulphur together brings about a
   1) Physical change 2) Chemical change
   3) Reversible change 4) Both physical and chemical change
6. Which of the following is not physical changes?
   1) Freezing of ice.
   2) Evaporation of water by the heat of sun.
   3) Expansion and contraction of metals on heating.
   4) Burning of paper
7. $Zn(OH)_2(s) \xrightarrow{heat} ZnO(s) + H_2O(g)$

   **$ZnO(s) \xrightarrow{heat} ZnO(s)$**

   **Cold and White** **Hot and yellow**

   1) Physical change 2) Chemical change
   3) Both physical and chemical change
   4) Neither physical nor chemical change
8. Choose the correct statements
   1) Heating of ammonium chloride strongly is a chemical change.
   2) Heating of Zinc oxide is a chemical change.
   3) Drying of a paint is a physical change.
   4) Evaporation of water is a physical change.

# ASSIGNMENT

1. Which of the following is an acid?
   1) Sea water 2) Tap water 3) Lime juice 4) Human saliva
2. Which of the following substances is an acid?
   1) Alcohol 2) Sea water 3) Ammonia 4) Soap solution
3. An acid will convert the colour of litmus paper to which colour?
   1) Red 2) Blue 3) Green 4) Yellow
4. The acids which are obtained from the minerals present in earth, are called:
   1) Organic acids 2) Strong acids 3) Inorganic acids 4)Weak acids
5. Which of the following acid is found in Olive Oil?
   1) Palmitic acid 2) Stearic acid 3) Oleic acid 4) Lactic acid
6. The formula of acetic acid is:
   1) $CH_2COOH$ 2) $CH_3COOH$ 3) $C_2H_4$ 4) $CH_3CHO$
7. The acids in which less than 30% of the molecules of it ionise in water to furnish $H^+$ ions, are called:
   1) Organic acids 2) Strong acids 3) Weak acids 4) Mineral acids
8. Choose the false statement:
   1) Sulphuric acid is a strong inorganic acid.
   2) Sulphurous acid is a weak inorganic acid.
   3) Nitrous acid is a weak inorganic acid.
   4) Phosphoric acid is a strong organic acid.
9. Which of the following acids is not present in acid rain?
   1) Carbonic acid 2) Sulphuric acid 3) Ascorbic acid 4) Nitric acid
10. Nitric acid is
    1) a strong aicd 2) a weak acid 3) a neutral acid 4) None
11. Addition of water
    1) Dilutes an acid 2) concentrates an acid 3) Both 4) None
12. Which of the following acids is called king of chemicals?
    1) Hydrochloric acid 2) Suphuric acid 3) Nitric acid 4) Formic acid
13. Which of the following acids is used in batteries?
    1) Hydrochloric acid 2) Suphuric acid 3) Nitric acid 4) Formic acid
14. Which of the following acids is used in TNT (tri nitro toluene)?
    1) Hydrochloric acid 2) Suphuric acid 3) Nitric acid 4) Formic acid
15. Vinegar is
    1) Formic aicd 2) Tartaric acid 3) Acetic acid 4) Oxalic acid
16. Which of the following acid is used as a presrvative?
    1) Formic aicd 2) Tartaric acid 3) Acetic acid 4) Oxalic acid

11. The focal length of the plane mirror is

1) At infinity 2) Zero 3) Negative 4) None

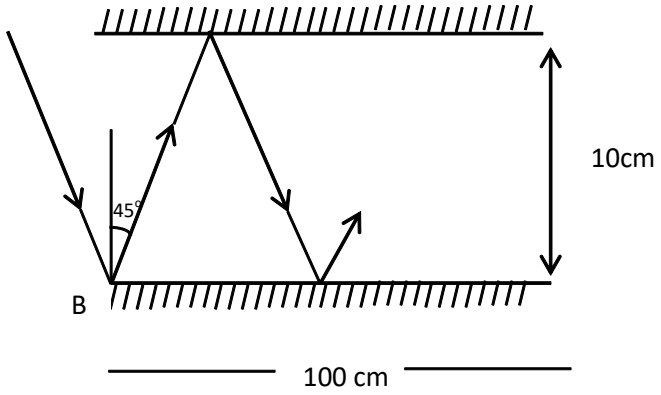
12. Objects that allow only some light to pass through them are considered _____.

1) transparent 2) translucent 3) opaque 4)solid

13. When a ray strikes a pane mirror at an angle of 15° with the mirror.What will be the angle through which ray get deviated

1)15° 2)30° 3)60° 4)150°

14. Two parallel plane mirrors A and B are placed at a seperation 10cm as shown in the figure .A ray incident on the corner of mirror B at an angle of incidence 45°.Find the number of times reflected from A




1)10 2)5 3)6 4)8

15. A plain mirror forms a virtual image. The distance between sangeetha and her image in a plane mirror is 20 m. How much distance should she move in order to get the distance of 30 m between herself and her image?

1) 2.5m away from the mirror 2) 2.5 m toward the mirror

3) 5m away from the mirror 4) 5 m towards the mirror

16. A boy stands 10 m in front of a plane mirror . then be moves 3 m towards the mirror. What is the distance between boy and his image

1) 20m 2) 14m 3) 7m 4) 6m

17. A boy is standing in front of a plane mirror at a distance of 3m from it. What is the distance between the body and his image?

1)3m 2) 6m 3)8m 4) 9m

8. A ___i___ mirror is used as a shaving mirror because when placed close to the face, it can form ___ii___ image of the face. Which alternative gives the correct information to complete the given statement?

| | (i) | (ii) |
|---|---|---|
| (A) | Convex | Diminished |
| (B) | Concave | Enlarged |
| (C) | Concave | Diminished |
| (D) | Convex | Enlarged |

1)A 2)B 3)C 4)D

9. A convex mirror tends to diverge a falling parallel beam of light. This property of convex mirrors is used in
1) Solar furnaces 2) Dentists' mirrors
3) Reflector in torch lights 4) Reflector in street lamps

10. There are four mirrors labelled as I, II, III, and IV. Mirrors I and IV can form virtual and small images only. Mirror III can form an image that can be obtained on a screen. Mirror II cannot form real images of objects. Which of the given mirrors can form an inverted image of the object?
1) I 2) II 3) III 4) IV

11. A child is standing in front of a magic mirror. She finds the image of her head bigger, the middle portion of her body of the same size, and of the legs smaller. The following is the order of combination for the magic mirror from the top:
1)Plane, convex and concave 2) Convex, Concave, and plane
3)Concave, plane, and convex 4)Convex, pane and concave

12. A small boy went to a mirror show and found that there were five mirrors. He went in front of each mirror to see his image. In the first four mirrors, his image appeared smaller in size and in the last mirror, his image appeared magnified. While his image was inverted in the first three mirrors, it was erect in the last two mirrors. How many convex mirrors were there in the mirror show?
1) 1 2) 2 3) 3 4) 4

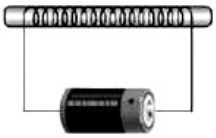
13. Convex mirrors are used as surveillance mirrors in libraries because they form
1) Real, erect, and magnified images
2) Real, erect, and diminished images
3) Virtual, erect, and magnified images
4) Virtual, erect, and diminished images

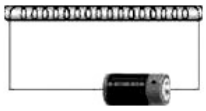
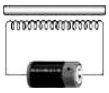
14. Which of the following mirrors are used by cosmetologists for their work?
1) Parabolic 2) Concave 3) Convex 4) Plane

15. While reading his science book, Billu decides to prepare a list of devices that involve the use of concave mirrors. The given table shows the list prepared by him. One device is listed incorrectly in the table.

Torch
Headlight
Dentist's mirror

17. Which of the following actions will produce a magnetic field?
    1) Rubbing cotton on a glass rod 2) Passing current through a wire
    3) Rubbing two iron bars together 4) Charging a conducting sphere
18. An electromagnet is not used in the functioning of which of the following devices?
    1) An electric motor 2) A generator 3) A telescope 4) A door bell
19. Suresh has an electromagnet in his school's electrical laboratory.



    Which of the following materials can be used to make the rod of an electromagnet?
    1) Glass 2) Wood 3) Iron 4) Plastic
20. John has to construct an electromagnet for his school project.
    Which of the following figures shows the way in which John should construct the electromagnet?
    1)  2) 3)  4)
21. The hair dryer works on the principle of __i__ effect of current. The component of the hair dryer that works on this principle is called __ii__ .
    The information in which alternative completes the given statements?

1)

| i | ii |
|---|---|
| heating | plate |

2)

| i | ii |
|---|---|
| magnetic | plate |

3)

| i | ii |
|---|---|
| magnetic | element |

4)

| i | ii |
|---|---|
| heating | element |

22. A pin is placed near a wire loop, as shown in the given figure. When current flows through the wire loop, the pin gets attracted toward the loop.

    Which force attracts the pin toward the loop?
    1) Electric 2) Magnetic 3) Frictional 4) Gravitational
23. In an electric bell, ____i____ energy is transformed into __ii__ energy.Which row completes the given statement?

1)

| i | ii |
|---|---|
| light | heat |

2)

| i | ii |
|---|---|
| electrical | heat |

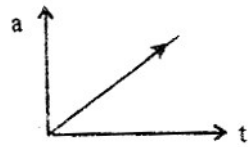
3)

| i | ii |
|---|---|
| electrical | sound |

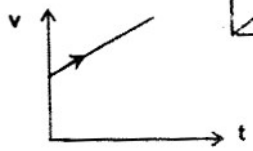
4)

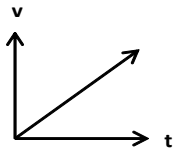
| i | ii |
|---|---|
| mechanical | light |

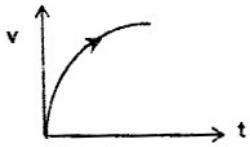
11. The acceleration-time graph of a particle moving in a straight line is shown in the figure-
The velocity time graph of the particle will be



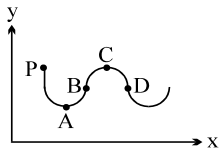

1)


2) 


3) 

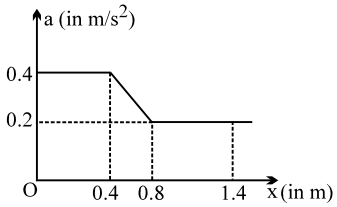

4) 


12. A man moves in x-y plane along the path shown. At what point is his average velocity vector in the same direction as his instantaneous velocity vector. The man starts from point P.



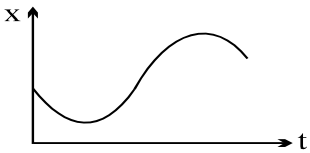

1) A 2) B 3) C 4) D

13. The acceleration of a particle which moves along the positive x-axis varies with its position as shown.
If the velocity of the rticle is 0.8 m/s at x = 0 , the velocity of the particle at x = 1.4 is(in m/s)



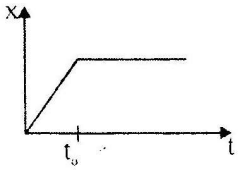

1) 1.6 2) 1.2
3) 1.4 4) none of these

14. The graph of position *x* versus time *t* represents the motion of a particle. If *b* and *c* are both positive constants, which of the following expressions best describes the acceleration a of the particle?




1) a = b – ct 2) a = + b
3) a = – c 4) a = b + ct

15. Fig. shows the displacement time graph of a particle moving on the X-axis.




1) the particle is continuously going in positive x direction
2) the particle is at rest
3) the velocity increases up to a time $t_0$, and then becomes constant.
4) the particle moves at a constant velocity up to a time $t_0$, and then stops.

**Q.No :16 -21**

The figure shows a velocity-time graph of a particle

# 7-LIGHT

## Introduction & Properties of Light

**1. Light**

If we enter a dark room, objects present there are not visible. However, if a bulb is switched on, everything in the room becomes visible. It shows that for vision the presence of light is essential.

**Definition of light:** It is an invisible energy which causes in us sensation of sight (vision). Since light is obtained from heat energy, i.e., when an object is heated to a temperature beyond [500 $^0$C, we can say that light is a kind of energy].

It must be kept in mind that light energy makes the surrounding objects visible, but by itself it is an invisible energy.

**2. Rectilinear propagation of light:**

Light travels in a straight line. This property of light is called rectilinear propagation of light.

**3. Reflection of light:** When a beam of light is incident on a surface, a part of it is returned back into the same medium. The part of light which is returned back into the same medium is called the reflected light.

**Regular and irregular reflection**:

**a) Regular Reflection:**

Reflection of light is of two kinds depending on the nature of the reflecting surface–regular reflection and diffused or irregular reflection.

When the reflecting surface is very well polished and smooth, the light that falls on it is regularly reflected. This phenomenon is known as regular refection.

**Note:** Regular reflection takes place on highly polished on smooth surfaces

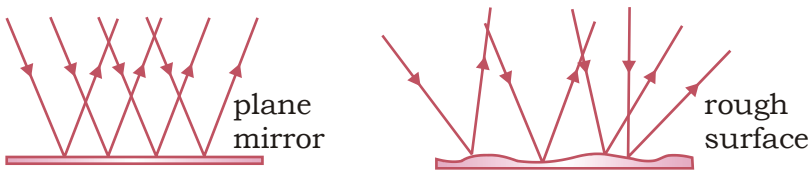
**Ex:** Plane mirror

b) **Irregular Reflection:**

If the reflecting surface is irregular, the rays of light that fall on it are scattered in all directions. This is called irregular or diffused reflection.

**Note:** Irregular reflection takes place on rough surfaces

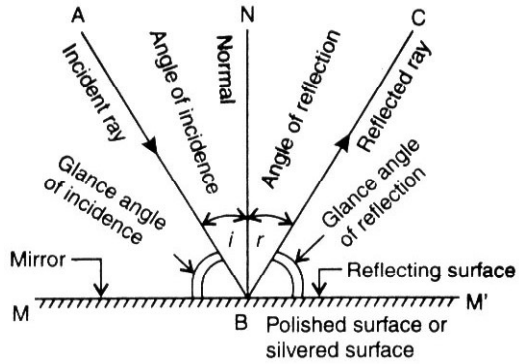
**Ex:** Wall, wood, paper, cinema screen, ground glass etc.




a. Regular reflection b. Irregular reflection

Two types of reflection

## 4. General terms to the Reflection




(a) **Mirror** : A smooth polished surface from which regular reflection can take place is called mirror. MM' is the mirror as shown in figure.

(b) **Incident Ray** : A ray of light which travels towards the mirror is called incident ray. AB is an incident ray in the figure.

(c) **Point of Incidence** : The point on the mirror, where an incident ray strikes is called point of incidence. 'B' is the point of incidence in the figure.

(d) **Reflected Ray** : A ray of light which bounces off the surface of a mirror, is called reflected ray. BC is reflected ray in the figure.

(e) **Normal**: The perpendicular drawn at the point of incidence, to the surface of mirror is called normal. BN is the normal in the figure.

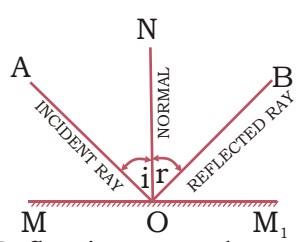
(f) **Angle of Incidence** : The angle made by the incident ray with the *normal* is called angle of incidence. $\angle$ABN is the angle of incidence in the figure.

(g) **Angle of Reflection** : The angle made by the reflected ray with the *normal* is called angle of reflection. $\angle$CBN is the angle of reflection in the figure.

(h) **Glance Angle of Incidence** : The angle which the incident ray makes with the **mirror** is called glance angle of incidence. $\angle$MBA is the glance angle of incidence in the figure.

(i) **Glance Angle of Reflection** : The angle which the reflected ray makes with the *mirror* is called glance angle of reflection. $\angle$M'BC is the glance angle of reflection in the figure.

**5. Laws of Reflection:** The reflection at a surface obeys the following two laws, which are called the laws of reflection.




Reflection at a plane surface

1. The angle of incidence 'i' is equal to the angle of reflection r (i.e. $\angle i = \angle r$).
   In figure $\angle$AON = $\angle$BON.
   For a ray incident normally on a surface, $\angle$ i = $0^0$, therefore $\angle$ r = $0^0$.

The information in which alternative completes the given statement?

1)

| i | ii |
|---|---|
| Land | day |

2)

| i | ii |
|---|---|
| Land | night |

3)

| i | ii |
|---|---|
| Sea | day |

4)

| i | ii |
|---|---|
| Sea | night |

13. __i__ is a good conductor of heat, but __ii__ is a bad conductor of heat.
The information in which alternative completes the given statement?

1)

| i | ii |
|---|---|
| Air | aluminium |

2)

| i | ii |
|---|---|
| Aluminium | air |

3)

| i | ii |
|---|---|
| Air | iron |

4)

| i | ii |
|---|---|
| Iron | aluminium |

14 When a layer of air receives enough heat from Earth's surface, it expands and becomes less dense. This layer is then pushed upward by buoyancy. The cool and heavy air sinks under this layer. This cycle repeats with the re-heating of the cool air.
The phenomenon described in the preamble is known as
1) Radiation 2) Induction 3) Conduction 4) Convection

15. Heat transfer by the process of __i__ requires medium while no medium is required for the heat transfer by the process of __ii__ .
The information in which alternative completes the given statement?

1)

| i | ii |
|---|---|
| radiation | convection |

2)

| i | ii |
|---|---|
| convection | conduction |

3)

| i | ii |
|---|---|
| conduction | convection |

4)

| i | ii |
|---|---|
| convection | radiation |

16. Which colour would you prefer to wear in summers?
1) White 2) Purple 3) Black 4) Red

17. Which of the following methods of heat transfer does not involve the movement or vibration of particles?
1) Radiation 2) Circulation 3) Convection 4) Conduction

18. Ravi kept some water for boiling for making tea. To mix sugar, he stirred it with a spoon. After some time, he observed that the temperature of the other end of the spoon had changed.
How did the temperature of the spoon change?
1) It became hot by the process of conduction.
2) It became cold by the process of conduction.
3) It became hot by the process of convection.
4) It became cold by the process of convection

19. Conduction, convection, and radiation are
1) Different modes of heat energy 2) Different modes of heat transfer
3) Different measures of temperature 4) All of these

20. The temperature of which Fahrenheit and reaumur scaleread the same is
1) –25.6 2) –20.6 3)25.6 4) 20.6

**Q: 21-23**

$$\frac{C}{5} = \frac{F-32}{9} = \frac{K-273}{5} = \frac{R}{4}$$

21. Express 100 K in °C
1) –273° 2) 100°C 3) 212°C 4) –173°C

# RAJESWARI BOOK PUBLISHERS

**NALLAKUNTA, HYDERABAD**

## FOUNDATION SCIENCE CLASS-7

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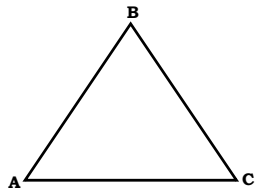
# 4-Distance , Displacement, Speed, Velocity

1. **Distance:-** The length of the path travelled by a body in certain interval of time is called distance.

   Suppose an object goes from A to C following path ABC, in a certain time (t). Then total distance travelled by object in time interval t = AB+BC




   If the object goes from A to B, B to C and C to A in time t, then the total distance travelled by object in time 't' is equal to AB+BC+CA

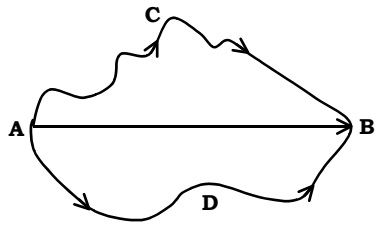
2. **Units of distance:-** In CGS system distance is measured in centimetres and in SI system it is measured in metres.

   Distance is denoted by the letter 'S'

   Distance is a scalar quantity. It's value can never be zero or negative, during the motion of an object.

3. **Displacement:-** The shortest distance between the initial position and the final position of a body is called displacement.

   Figure shows two points A and B which can be reached by the paths AB, ACB or ADB or any other supposed path.




   However, the shortest distance between two points A and B is AB. Thus, AB is the displacement of a body.

   Displacement is a vector quantity, and hence, is directional. Without direction, displacement has no meaning. For example, if a body moves 10m, it means the body has covered a distance of 10 m in any direction. However, if a body moves 10 m towards west, then displacement of the body is 10m (west).

4. **Characteristics of displacement:-**

   1. The displacement of an object has the unit of length.
   2. The displacement of an object in a given interval of time can be positive, zero or negative.
   3. The magnitude of the displacement of an object between two points gives the shortest distance between those two points.
   4. The displacement of the object between two points has a unique value.
   5. The actual distance travelled by the object in the given time interval can be equal to or greater than the magnitude of the displacement.